A Unique Stationary Equilibrium at Relative Risk Aversion Three: How the Result Is Built

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Abstract

In a life-cycle economy with uninsurable earnings risk, a borrowing constraint, sixty-period lives and relative risk aversion three, at one rational calibration, there is exactly one stationary equilibrium interest rate in $(-8\%, 6\%]$, and it lies in $(4\%, 6\%)$. The route is single crossing in derivative form: excess demand for capital is strictly increasing on the bracket because the firm's demand falls faster than household supply can, and a certified lower bound on the slope of supply, built from rigorous enclosures of the sixty consumption functions, sits at most a third of the way to the slope of demand. The note states the results the construction rests on — the stage properties of the finite backward induction, the propensity floor, the cohort envelope, continuity of supply, the interval Euler step — and then explains how they are assembled: the derivative of consumption in the rate from the Euler equality, the interval recursion that bounds it on grid intervals, the frozen-transport argument that carries a bound from a rate to a cell of rates by continuation, and the three hundred and seventy cells that cover $[2.3\%, 6\%]$; below 2.3% Lean theorems take over. Every step but the arithmetic is a theorem; the arithmetic is exact rational arithmetic rounded outward, evaluated by a program rather than replayed in the kernel. The note closes with what that standing is and what would raise it.

1 Introduction

The companion write-up proves that the stationary equilibrium of a life-cycle Bewley economy is unique at logarithmic utility, by Light's three theorems in the life-cycle setting: saving rises with the interest rate at every state (Theorem 1), ordered policies give ordered capital supply (Theorem 2), and a rising supply meets the firm's falling demand at most once (Theorem 3). Above logarithmic utility Theorem 1 is false in fact — at relative risk aversion five it fails

*The Lean development described here is at `https://github.com/LeanEconomics/LeanEconomics`. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington). This note is a self-contained account of one result of the overlapping-generations write-up in the same repository, *Overlapping Generations in Lean: The Life-Cycle Household*, which it cites as the companion; the results it builds on are stated with their formal names and not reproved. It is a project record, not a paper for submission.

pointwise at the youngest ages in the top earnings states — and no analytic route around it is known. What this note records is the result that reaches $\gamma = 3$ anyway, and the construction that reaches it.

The result is a statement about one economy at one calibration. Fix sixty-period lives, a seven-state earnings chain, $\beta = 24/25$, $\gamma = 3$, and a Cobb–Douglas firm. Then on the range $(-8\%, 6\%]$ there is exactly one stationary equilibrium, and it lies in $(4\%, 6\%)$. The argument is single crossing, but not in the form Theorem 1 supplies. Single crossing needs only that excess demand $E(r) = S(r) - D(r)$, supply minus demand for capital, be strictly increasing; and since demand falls steeply at the calibration, supply may fall too, as long as it falls more slowly. The construction certifies a lower bound on the slope of supply, $S'(r) \geq -B_d$ with $B_d \leq 9.2$, against a demand slope of magnitude at least 28.7, everywhere on the bracket; and it certifies that supply is below demand at 4% and above it at 6%. Continuity of supply, which is a theorem, then gives an equilibrium in the bracket, and strict monotonicity of E gives at most one.

The word *certifies* carries the standing of the result and the note is careful with it. Three kinds of statement appear. Theorems checked by the Lean kernel, which are stated with their formal names and not reproved: these are the companion's. Paper arguments, which are short and are given in full here: the derivative of the Euler equality, the mean-value link from a derivative bound to a secant bound, and a continuation argument. And arithmetic, which is exact rational arithmetic rounded outward, carried out by a program; its inequalities are instances of the formalised interval Euler step and of the recursions this note writes down, but the program that evaluates them has not been replayed in the kernel. Section 13 tabulates which step is which.

The note is organised as a construction. Section 2 fixes the economy and states the theorem. Section 3 states the known results. Section 4 reduces uniqueness to a bound on the slope of supply, Section 5 derives the derivative of consumption in the rate and the envelope that turns its failure into a slope bound, Section 6 describes the enclosures of the consumption functions, Section 7 the interval recursion that bounds the derivative, Section 8 the passage from a rate to a cell of rates, Section 9 the covering of the bracket, Section 10 the endpoint certificate, and Section 11 assembles the proof. Section 13 is about standing and what remains.

2 The economy and the statement

2.1 Households

A household lives $J = 60$ periods, is born with no assets, and dies leaving nothing. Earnings $y(z)$ follow a finite Markov chain with transition matrix π on $n_z = 7$ states; the gross return on assets is $R = 1 + r$; cash on hand is $m = Ra + y(z)$; saving a' is bounded below by a zero borrowing limit and above by an artefactual cap $\bar{a}$; utility is CRRA, $u'(c) = c^{-\gamma}$, with $\gamma = 3$ and discount factor β. Periods are counted from the end of life. With k periods still to come

after the present one, the continuation value is the k-th Bellman iterate from zero,

$$V_k = T^k 0, \qquad (Tv)(a,z) = \max_{a' \in [0,\bar{a}]} \Big\{ u(m-a') + \beta \sum_{z'} \pi(z,z')\, v(a',z') \Big\},$$

the stage-k policy is $g_k = \text{policyOf}\, V_k$ and the stage-k consumption is $c_k = m - g_k$ (`stageValue`, `stagePolicy`, `stageConsumption`). Age j of life is stage $k = J - 1 - j$; in the last period the household consumes its cash on hand, $c_0 = m$. The problem is a finite backward induction, so every property of the stage policies is proved by induction on k with an explicit base case, and nothing is a fixed point.

2.2 Earnings

The chain is the Rouwenhorst discretisation with persistence $\rho = 9/10$ on seven states. Its invariant law is binomial, $\nu = (1,6,15,20,15,6,1)/64$, and the earnings levels are the seven values $(0.301, 0.449, 0.670, 1, 1.492, 2.226, 3.320)$ divided by their ν-mean 1.1266, so that mean earnings are one:

$$y = (0.2672,\ 0.3986,\ 0.5947,\ 0.8876,\ 1.3244,\ 1.9759,\ 2.9470).$$

All of these are rationals, and so is every quantity the certificate computes.

2.3 The firm and the equilibrium

Technology is Cobb–Douglas with capital share $\alpha = 9/25$ and depreciation $\delta = 2/25$. In wage units — capital per unit of the wage, the normalisation homotheticity allows — the firm's demand for capital at rate r is

$$D(r) = \frac{\alpha}{(1-\alpha)(r+\delta)}, \qquad D'(r) = -\frac{\alpha}{(1-\alpha)(r+\delta)^2},$$

so $D(4\%) = 4.6875$, $D(6\%) = 4.0179$, and $|D'|$ falls from 39.06 at 4% to 28.70 at 6%.

Every cohort is born with no assets and an earnings state drawn from ν; cohorts have equal weight. Write $(\mathcal{S}_k h)(a,z) = \sum_{z'} \pi(z,z')\, h(g_k(a,z),z')$ for the expectation of h next period (`stageStep`). The assets a household holds over the rest of its life, seen from its current state, satisfy

$$A_0(a,z) = a, \qquad A_{k+1}(a,z) = a + (\mathcal{S}_{k+1} A_k)(a,z)$$

(`cohortAssets`), and capital supply per head is

$$S(r) = \frac{1}{J} \sum_{z} \nu(z)\, A_{J-1}(0,z)$$

(`olgCapital`): a finite sum of finite sums, with no stationary distribution anywhere. A stationary equilibrium is a rate at which $S(r) = D(r)$.

2.4 The statement

Theorem 2.1 (Uniqueness on the bracket at $\gamma = 3$)**.** *In the economy above, with $J = 60$, $\beta = 24/25$, $\gamma = 3$, the seven-state chain, $\alpha = 9/25$ and $\delta = 2/25$, there is exactly one stationary equilibrium interest rate in $(-8\%, 6\%]$, and it lies in $(4\%, 6\%)$. Specifically: S is continuous, $S(4\%) < D(4\%)$ and $S(6\%) > D(6\%)$, so an equilibrium exists in $(4\%, 6\%)$; $E = S - D$ is strictly increasing on $[2.3\%, 6\%]$, so there is at most one there; and $S < D$ on $(-8\%, 2.3\%]$ (Theorem 12.1), so there is none there.*

The theorem is silent above 6%. Theorem 12.1 excludes $[8.7\%, 20\%]$ by proof term, so $(6\%, 8.7\%]$ is the only range below 20% in which an equilibrium is not yet excluded; closing it is machine time, not new argument, and Section 13 returns to it.

3 The results the construction rests on

Each result below is a theorem of the companion development, checked by the Lean kernel, and is used here as a black box. The formal name follows each statement.

Stage properties. At every stage k, every state and every asset level in the region: consumption is positive (`stageConsumption_pos`); saving and consumption are both nondecreasing in assets (`stagePolicy_mono`, `stageConsumption_mono`); consumption is concave in assets (`concaveOn_stageConsumption`, Carroll–Kimball); and the two Euler inequalities hold, under the cap,

$$\beta R \sum_{z'} \pi(z, z')\, u'\big(c_k(g_{k+1}(a, z), z')\big) \;\le\; u'\big(c_{k+1}(a, z)\big),$$

at every state (`stage_euler_le`), and the reverse inequality at every state where the household saves (`stage_euler_ge`). Where the household saves, the two together are the Euler *equality*.

The propensity floor. Let $\text{þ} = (\beta R)^{1/\gamma}$ and define the finite-horizon minimal propensities to consume by $\kappa_0 = 1$, $\kappa_{k+1} = \kappa_k R/(\text{þ} + \kappa_k R)$, so that $1/\kappa_k = \sum_{i \le k} (\text{þ}/R)^i$ (`stageMPC`). Then $c_k(a, z) \ge \kappa_k m$ (`stageMPC_mul_le_stageConsumption`), and, the sharper statement that the argument needs, the propensity to consume out of assets is at least κ_k:

$$\kappa_k R\,(a_2 - a_1) \;\le\; c_k(a_2, z) - c_k(a_1, z), \qquad a_1 \le a_2, \tag{1}$$

so saving is Lipschitz in assets with constant $(1 - \kappa_k)R$ (`stageMPC_mul_le_stageConsumption_sub`). A lower bound on κ_k follows from any upper bound $t \ge \text{þ}$ as $1/\sum_{i \le k} (t/R)^i$ (`stageMPC_ge_of_patience_le`).

The cohort slope. With $G_0 = 1$ and $G_{k+1} = 1 + G_k(1 - \kappa_{k+1})R$ (`cohortSlope`), accumulated assets are Lipschitz in assets, $A_k(a_2, z) - A_k(a_1, z) \le G_k(a_2 - a_1)$ for $a_1 \le a_2$ in the region (`cohortAssets_sub_le_cohortSlope`).

The earnings marginal. Since ν is invariant for π and the cohort is born from ν, the ν-weighted expectation of any function of the earnings state is unchanged by any number of steps of the cohort recursion (`sum_mul_incomeStep_iterate`). An expectation of a failure over the cohort is therefore bounded by $\sum_z \nu(z)\, \sup_a(\text{failure at } (a,z))$: exact in earnings, worst-case only in assets.

Theorem 1 is an affine bound. Since $a' = Ra + y(z) - c$ at either rate, for $r_1 < r_2$,

$$g_k(a,z;r_2) \geq g_k(a,z;r_1) \quad \Longleftrightarrow \quad c_k(a,z;r_2) - c_k(a,z;r_1) \leq (r_2 - r_1)\, a \tag{2}$$

(`stagePolicy_le_iff_consumption_le`). The failure of Theorem 1 at a state is the positive part of the excess of the left side over the right.

The cohort envelope. Couple the two economies on a common earnings path from the common start $a_0 = 0$. If, at every age, the failure of Theorem 1 at the lower-rate state is at most e_j in ν-expectation and the higher-rate saving is Lipschitz with constant L_j in assets, then

$$S(r_2) - S(r_1) \;\geq\; -\frac{1}{J}\sum_j f_j, \qquad f_0 = 0, \quad f_{j+1} = L_j f_j + e_j, \tag{3}$$

which in cohort form, with the regions of the lower-rate economy and G_k as the accumulated Lipschitz constants, is `olgCapital_sub_ge_of_failure`. The pointwise ordering of the policies is not assumed; its failure is priced.

Continuity, existence, single crossing. Capital supply is continuous in the rate (`continuousOn_olgSupply`); if it lies below a continuous demand at the bottom of an interval and above it at the top, the two meet (`exists_olgEquilibrium`); and a supply that never falls meets a strictly falling demand at most once (`olgEquilibriumRate_unique`). The last is used here in its elementary form — a strictly increasing E has at most one zero — which needs no theorem.

Reachable regions. The artefactual cap must be slack where the household can be. A family of asset sets, one per stage, each inside $[0, \bar{a}]$, each carried by the next stage's policy into the previous set, and on each of which the policy stays strictly below the cap, is a reachable family (`StageRegions`); a cohort born with nothing stays in its family (`reachRegions`), and every stage property above holds on any reachable family. Below, the family is the enclosure's own reach: the sets $[0, \rho_j]$ with ρ_j the certified maximal assets at age j, about 44 at most, far below any cap the computation uses.

The interval Euler step. Let $E_k(x) = \beta R \sum_{z'} \pi(z,z')\, x(z')^{-\gamma}$ and $m = y_z + Ra$.

- If $0 < c_L \leq m$ and $E_k(\ell) \leq c_L^{-\gamma}$ for numbers $\ell(z') \leq c_k(m - c_L, z')$, then $c_L \leq c_{k+1}(a,z)$ (`le_stageConsumption_of_cert`).

- If $0 < c_H \le m$, $m - c_H$ lies in the region, and $c_H^{-\gamma} \le E_k(h)$ for numbers $h(z') \ge c_k(m - c_H, z')$, then $c_{k+1}(a, z) \le c_H$ (`stageConsumption_le_of_cert`).
- If $E_k(\ell) \le m^{-\gamma}$ for $\ell(z') \le c_k(0, z')$, then $c_{k+1}(a, z) = m$: the household is at the corner (`stageConsumption_eq_resources_of_cert`).
- Between grid points a concave nondecreasing function lies above the chord of lower bounds at two points and below the left secant extended forward and the right secant extended backward (`le_of_chord`, `le_of_left_secant`, `le_of_right_secant`).

These are the only inequalities the enclosure of Section 6 ever asserts.

4 Uniqueness as a bound on the slope of supply

Single crossing asks that $E = S - D$ be strictly increasing on the bracket. Since $D'(r) \le -28.7$ on $[2.3\%, 6\%]$, that holds as soon as supply does not fall faster than demand:

$$S'(r) \;>\; D'(r) \quad \text{on } [2.3\%, 6\%], \qquad \text{equivalently} \qquad -S'(r) \;<\; |D'(r)|. \tag{4}$$

Two remarks fix what (4) means and what it does not need.

First, S is not differentiable everywhere. Each stage policy has a kink at the asset level where the borrowing constraint stops binding; kinks propagate backward to wherever a policy sends a household onto a kink of the next stage; and S, a finite composition of the policies along the newborn's paths, inherits finitely many kinks in r. Off the kinks every policy is real-analytic in (a, r) by the analytic implicit function theorem applied to the Euler equality, so S is continuous and piecewise C^1 with finitely many pieces, each extending C^1 to its closed piece. A continuous, piecewise-C^1 function with positive derivative on each open piece is strictly increasing. So (4), read on the open pieces, suffices. (The companion's finiteness theorem, `PiecewiseAnalyticOn.finite_zeros`, is the same observation one level down: piecewise analytic with no flat piece gives finitely many zeros. It is not needed here, because (4) gives the stronger conclusion directly.)

Second, the true slope of supply is large and positive — about $+53$ at 4% and $+62$ at 6% on the solved model — so (4) holds in fact with room to spare, and a proof of it may be lossy by more than the true slope and still close. The construction below certifies $S' \ge -B_d$ with B_d between 6.5 and 9.2: a bound that does not even know supply rises, and is still a third of what is needed.

The link from a derivative bound to the secant inequality that single crossing is usually stated with is the mean value theorem, and it is worth writing down because it is the step that the formalised envelope (3) does not take.

Lemma 4.1 (From the derivative to the secant)**.** *Let $r_1 < r_2$ lie in a cell on which, for every stage k, every earnings state and every asset level in the stage's region, $c_k(a, z; \cdot)$ is continuous and piecewise C^1 in the rate with finitely many pieces, and suppose*

$$\big(\partial_r c_k(a, z; r) - a\big)^+ \;\le\; \varepsilon_k(a, z) \qquad \textit{at every } r \textit{ in the cell off the kinks.}$$

Then the failure of Theorem 1 between r_1 and r_2 at (a,z) is at most $(r_2-r_1)\,\varepsilon_k(a,z)$, and the envelope (3) *gives*

$$\frac{S(r_2)-S(r_1)}{r_2-r_1} \;\geq\; -\frac{1}{J}\sum_k G_k \sum_z \nu(z) \max_{a\in reach} \varepsilon_k(a,z) \;=:\; -B_d.$$

Proof. A continuous, piecewise-C^1 function is absolutely continuous, so $c_k(a,z;r_2) - c_k(a,z;r_1) - (r_2-r_1)a \;=\; \int_{r_1}^{r_2}(\partial_r c_k - a)\,dr \;\leq\; (r_2-r_1)\varepsilon_k(a,z)$, and the positive part is bounded by the same. Insert $e_j = (r_2-r_1)\sum_z \nu(z)\max_a \varepsilon$ and $L_j = (1-\kappa_k)R_2$ into (3); the accumulation f_j is linear in the e's, and $\sum_j f_j$ with the L_j above is $\sum_k G_k e$ by the definition of G, with R_2 in place of R. Divide by r_2-r_1. □

So the whole problem is to certify, on every cell of a partition of the bracket, a bound $\varepsilon_k(a,z)$ on the *derivative-form failure* $(\partial_r c_k - a)^+$ at every state in the cohort's reach, and to check $B_d < \min_{\text{cell}} |D'|$. The rest of the note is that certificate.

5 The derivative of consumption in the rate

5.1 The formula

Fix a stage k, an earnings state z and an asset level a at which the household saves, so that the Euler equality holds:

$$c^{-\gamma} = \beta R \sum_{z'} \pi(z,z')\, c_k(x,z')^{-\gamma}, \qquad x = m - c = Ra + y(z) - c,$$

with $c = c_{k+1}(a,z)$. Differentiate in r at fixed a, using $\partial_r R = 1$ and $\partial_r x = a - \partial_r c$:

$$-\gamma c^{-\gamma-1}\,\partial_r c = \beta \sum_{z'} \pi\, c_k^{-\gamma} - \gamma\beta R \sum_{z'} \pi\, c_k^{-\gamma-1}\big[\partial_x c_k\,(a-\partial_r c) + \partial_r c_k\big].$$

Multiply by $-c^{\gamma+1}/\gamma$ and use the Euler equality on the first term, $\beta c^{\gamma+1}\sum \pi c_k^{-\gamma} = c/R$. With the *Euler weights*

$$w_{z'} = \beta R\,\pi(z,z')\,c_k(x,z')^{-\gamma-1}\,c^{\gamma+1}, \qquad A = \sum_{z'} w_{z'}, \qquad \Psi_x = \sum_{z'} w_{z'}\,\partial_x c_k(x,z'), \quad (5)$$

this is $\partial_r c = -c/(\gamma R) + \Psi_x(a-\partial_r c) + \sum_{z'} w_{z'}\partial_r c_k$, hence

$$\partial_r c_{k+1}(a,z) = \frac{-\dfrac{c}{\gamma R} + \Psi_x\, a + \sum_{z'} w_{z'}\,\partial_r c_k(x,z')}{1+\Psi_x}, \qquad \partial_r g_{k+1} = a - \partial_r c_{k+1}. \quad (6)$$

At a corner, $c = m$ and $\partial_r c = a$ exactly. In the last period, $c_0 = m$ and $\partial_r c_0 = a$ everywhere; that is the terminal condition of a backward recursion in which (6) is the step.

The three ingredients of the step are bounded by things the construction controls. The weights $w_{z'}$ are built from consumption at two consecutive stages; A is the power-mean amplification of the companion's Section 7, close to one. The slope $\partial_x c_k$ of next period's

consumption in next period's assets lies in $[\kappa_k R, R]$ by the propensity floor (1) and monotone saving, so $\Psi_x \in [\kappa_k R\,A,\ R\,A]$. And $\partial_r c_k$ is the previous step's output.

5.2 The derivative-form failure and the three recursions

The failure of Theorem 1 in derivative form is

$$e_{k+1}(a,z) = \left(\partial_r c_{k+1}(a,z) - a\right)^+ = \left(-\partial_r g_{k+1}(a,z)\right)^+,$$

the amount by which saving falls with the rate. Rewriting (6),

$$\partial_r c_{k+1} - a = \frac{-\dfrac{c}{\gamma R} - a + \sum_{z'} w_{z'}\,\partial_r c_k(x,z')}{1+\Psi_x},$$

so the failure at stage $k+1$ is driven by $\sum_{z'} w_{z'}\partial_r c_k$ exceeding $a + c/(\gamma R)$, damped by $1+\Psi_x$. Upper and lower bounds $U_k \geq \partial_r c_k \geq L_k$ propagate through (6) from $U_0 = L_0 = a$: the numerator's upper bound uses U_k where $w_{z'} > 0$, the smallest admissible c and the largest admissible a; and the denominator's range is the range of Ψ_x.

How much is kept of (6) decides whether the bound is useful, and the companion measured three versions on the solved model. The *crude* recursion replaces $\sum w\,\partial_r c_k$ by $A \max_{z'} U_k$ and Ψ_x by its range; it is what a proof from the two Euler inequalities alone would deliver, and it is dead — $B_d = 312$ at $\gamma = 3$ against 39 needed, its worst state the newborn at zero wealth. The *weighted* recursion keeps $\sum_{z'} w_{z'} U_k$ with the Euler weights and asks nothing of Ψ_x beyond its range; it gives $B_d \approx 4$, a margin of ten. The *exact* recursion keeps Ψ_x too and is the chain rule. What separates the crude from the weighted recursion is a single max over tomorrow's states, and that is the lesson the certificate is built on: the weights $w_{z'}$ must be carried, which means the consumption functions at two consecutive stages must be enclosed tightly enough that the weights are known to a few percent. The companion's sensitivity check — every weight, slope and interpolated bound perturbed adversarially by one percent — leaves the weighted bound at about 4.2, so enclosures of that quality suffice.

6 Enclosing the consumption functions

6.1 The endogenous-grid recursion, in intervals

The enclosure is the endogenous-grid recursion of the Euler equality, run in integers over $D = 10^{15}$ with every operation rounded outward (`numerics/rational/encl.py`, shared by `derivcert.py`). At stage $k+1$, state z and asset level a with $m = y_z + Ra$, consumption is m where $\Psi_k(0) \geq m$ and otherwise the unique solution of $c = \Psi_k(m-c)$, where

$$\Psi_k(a') = \Big[\beta R \sum_{z'} \pi(z,z')\,c_k(a',z')^{-\gamma}\Big]^{-1/\gamma}$$

is the consumption today that makes the Euler equality hold against saving a'. Ψ_k is increasing in a' (monotone consumption) and increasing in c_k, so bracketing c_k brackets Ψ_k: from stage-k bounds $\underline{c}_k \le c_k \le \bar{c}_k$ on the grid, build Ψ_k^{lo} from $\underline{c}_k$ with every rounding downward and Ψ_k^{hi} from $\bar{c}_k$ with every rounding upward. Since $c \mapsto \Psi(m-c) - c$ is strictly decreasing,

$$c_L \le \Psi_k^{\mathrm{lo}}(m - c_L) \;\Rightarrow\; c_L \le c_{k+1}(a,z), \qquad c_H \ge \Psi_k^{\mathrm{hi}}(m - c_H) \;\Rightarrow\; c_H \ge c_{k+1}(a,z),$$

which are exactly the first two clauses of the interval Euler step, with $\ell = \underline{c}_k$ and $h = \bar{c}_k$ read at $m - c_L$ and $m - c_H$. Each certified bound costs one rigorous evaluation of Ψ from a floating-point candidate; if the candidate fails, one rigorous iteration of Ψ is itself certified (since Ψ^{lo} is nondecreasing, $\Psi^{\mathrm{lo}}(m-t)$ for $t \ge c^*$ is a certified lower bound), and after six iterations a bisection in rational arithmetic is the fallback. The corner region $[0, a^*]$ on which $c = m$ exactly is certified at each stage and state from $\Psi_k^{\mathrm{lo}}(0)$: wherever $y_z + Ra \le \Psi_k^{\mathrm{lo}}(0)$, the third clause of the interval Euler step applies. The terminal stage is exact, $c_0 = m$.

6.2 Between grid points

Consumption is enclosed at the grid points; the recursion needs it at the saving $m - c$, which is not a grid point. Monotonicity and concavity close the gap: on $[a_i, a_{i+1}]$ the function lies above the chord of the two lower bounds, and below each of three upper bounds — the right endpoint's upper bound (monotonicity), the left secant through the bounds at a_{i-1}, a_i extended forward, and the right secant through the bounds at a_{i+1}, a_{i+2} extended backward — of which the smallest is taken. Above the certified corner the chord is anchored at the corner itself, where consumption is known exactly. These are the last clause of the interval Euler step.

6.3 The grid, and the widths

The grid has 1200 points from 0 to 3 in steps of 0.0025 and then relative steps of 0.25% up to 3000: 3968 points in all. The top is far above the reach of any cohort (about 44) because unavailability beyond the grid's top creeps inward by a factor of about $1/R$ per stage and must never reach the cohort's region; the fine spacing is for the derivative recursion, whose interpolation slack is of order $c''h^2$ per stage accumulated over a transient of some fifty stages. Two details of the implementation mattered for the widths. The floating-point candidate for each certified fixed point must be computed on its own side of the enclosure — the lower candidate from Ψ^{lo}, the upper from Ψ^{hi} — or each stage loses a fraction Ψ_x of the width; and the corner test must be strict (the corner at zero wealth is certified only where the household demonstrably does not save there). With both, the enclosure width at the newborn stage over the cohort's region is 0.0023 at 4% and 0.0015 at 6%, against consumption of order one, which is what the weights need.

What the enclosure delivers at a single rate is, at every stage k, every state z and every grid point a_i, certified numbers $\underline{c}_k(a_i,z) \le c_k(a_i,z) \le \bar{c}_k(a_i,z)$ and a certified corner $a_k^*(z)$, each one an instance of the interval Euler step; and from these, by the chord and secants,

bounds at every asset level up to the grid's top. The cohort's reach ρ_j is computed from the same data: $\rho_0 = 0$ and $\rho_{j+1} = \max_z(m(\rho_j) - \underline{c}_{J-1-j}(\rho_j, z))$, the most any household can hold at age $j+1$, since saving is nondecreasing in assets.

7 The interval recursion for the derivative

A derivative has no monotonicity or concavity to interpolate with, so the bounds on $\partial_r c_k$ are carried on grid *intervals* rather than at grid points: $U_k[z][i] \geq \partial_r c_k(a, z) \geq L_k[z][i]$ for every $a \in [a_i, a_{i+1}]$. The terminal values are $U_0[z][i] = a_{i+1}$ and $L_0[z][i] = a_i$, since $\partial_r c_0 = a$. One step, from stage k to stage $k+1$ on the interval $[a_i, a_{i+1}]$ at state z, evaluates (6) with every ingredient replaced by its range over the interval and rounded outward:

- today's consumption $c \in [\underline{c}_{k+1}(a_i, z), \overline{c}_{k+1}(a_{i+1}, z)]$, by monotonicity;
- today's saving $x = m - c \in [m(a_i) - \overline{c}_{k+1}(a_i, z),\ m(a_{i+1}) - \underline{c}_{k+1}(a_{i+1}, z)]$, since saving is nondecreasing in assets;
- tomorrow's consumption $c_k(x, z')$ in the enclosure at the two ends of that x-range, by monotonicity;
- each weight $w_{z'}$ between its two extreme values over those ranges, and Ψ_x in $[\kappa_k R\, A,\ R\, A]$ with $A = \sum w$ and κ_k from its certified lower bound;
- $\partial_r c_k(x, z')$ between the smallest L_k and the largest U_k over the grid intervals that the x-range meets.

The interior formula needs the Euler equality, so it is used only where saving is positive on the whole interval, $m(a_i) - \overline{c}_{k+1}(a_i, z) > 0$. On every other interval — those inside the certified corner, where $\partial_r c = a$ exactly, and those that straddle the point where the constraint stops binding, where the household may be at the corner at some a in the interval — the bounds are widened to cover $\partial_r c = a$ as well: $U \leftarrow \max(U, a_{i+1})$, $L \leftarrow \min(L, a_i)$. Beyond the grid's top no bound is available, and that unavailability creeps inward from 3000 by about one interval per stage, never approaching the reach.

The failure bound on the interval is $\varepsilon_{k+1}[z][i] = (U_{k+1}[z][i] - a_i)^+$, the largest value $\partial_r c - a$ can take there. By Lemma 4.1 what enters the envelope is, for each age j (stage $J-j$), $\sum_z \nu(z) \max\{\varepsilon_{J-j}[z][i] : a_i \leq \rho_{j-1}\}$, the maximum over the intervals inside the reach; and then

$$B_d = \frac{1}{J}\sum_{j=1}^{J} f_j, \qquad f_1 = 0, \quad f_{j+1} = (1 - \kappa_{J-j}) R\, f_j + \sum_z \nu(z) \max_{\text{reach}} \varepsilon_{J-j}[z][\cdot],$$

with κ its certified lower bound (which makes $1 - \kappa$ an upper bound) and R the gross rate. Every number in this display is a rational, computed exactly.

At a single rate on the fine grid this gives, for $\gamma = 3$, $B_d = 5.2$ at 4% against $|D'| = 39.1$ and $B_d = 7.4$ at 6% against 28.7; and for $\gamma = 5$, $B_d = 12.8$ at 4%. The failure is concentrated at the youngest ages — about 0.4 per age at ages one to ten, 0.23 at age thirty — which is

where the enclosures are widest and where, on the solved model, the true derivative-form failure is exactly zero at $\gamma = 3$: the bound is interpolation slack, not economics. That is why the grid is fine.

8 From a rate to a cell of rates

Lemma 4.1 needs the derivative-form failure bounded at every rate in the cell, so the enclosures the recursion reads must hold at every rate in the cell. A direct cell enclosure — the same endogenous-grid recursion with the two rates at the ends of the cell placed where each hurts — is valid but too loose: its widths are twenty to seventy times the single-rate widths on cells of a tenth of a percentage point, and the recursion's weights carry consumption to the power $\gamma + 1$. A cell enclosure fed back from the recursion's own derivative bound diverges for the same reason. What works is to *freeze* the transport and verify it afterwards.

8.1 The frozen transport

Let h be the cell's width and r_0 its midpoint. Run the point enclosure at r_0 to get $[\underline{c}_k^0, \overline{c}_k^0]$ at every stage, state and grid point, and run the derivative recursion at r_0 to get U_k^0. Now *assume* that on the cell, at every stage and every state on a wealth zone Z_k to be specified,

$$|\partial_r c_k(a, z; r)| \;\le\; M_k[z][i] := \lambda \max(|U_k^0[z][i]|, |L_k^0[z][i]|) + \mu, \qquad a \in [a_i, a_{i+1}] \subset Z_k, \quad (7)$$

with $\lambda = 2$ and $\mu = 1$. Under the assumption, consumption at any rate in the cell is within $M_k h/2$ of its value at the midpoint, so the transported enclosure

$$\underline{c}_k(a_i, z) = \underline{c}_k^0(a_i, z) - M_k h/2, \qquad \overline{c}_k(a_i, z) = \overline{c}_k^0(a_i, z) + M_k h/2$$

holds on the whole cell on the zone. Outside the zone the enclosure falls back to bounds that need no assumption at all: the propensity floor $\kappa_k m$ from below, read at the lower rate, and cash on hand m from above, read at the upper rate. Then run the derivative recursion on the transported enclosures, with every rate-dependent quantity taken at whichever end of the cell makes the bound larger, to get bounds U_k valid at every rate in the cell *at which the transported enclosure holds*, and check, on every interval where (7) was assumed,

$$\max(|U_k[z][i]|, |L_k[z][i]|) \;<\; M_k[z][i] \qquad \text{strictly.} \quad (8)$$

Lemma 8.1 (Continuation)**.** *Suppose that for every stage k, state z and asset level a in the cohort's region, $r \mapsto c_k(a, z; r)$ is continuous and piecewise C^1 on the cell with finitely many pieces, the derivative extending continuously to each closed piece; and that the derivative recursion is sound: whenever the transported enclosure holds at a rate r in the cell on the zone, and the fallback bounds outside it, the recursion's output satisfies $L_k \le \partial_r c_k(\cdot; r) \le U_k$ on every interval, one-sidedly at kinks. If* (8) *holds wherever* (7) *was assumed, then* (7) *is true on the whole cell, and so the transported enclosure and the bounds U_k, L_k hold at every*

rate in the cell.

Proof. Let T be the set of $t \in [0, h/2]$ such that $|c_k(a, z; r) - c_k(a, z; r_0)| \leq M_k|r - r_0|$ for all $|r - r_0| \leq t$, all stages, states and asset levels in the zone. T contains 0, and is closed because each inequality is non-strict and c_k is continuous in r. Let $t \in T$ with $t < h/2$. For every r with $|r - r_0| \leq t$ the transported enclosure holds (the midpoint enclosure holds at r_0, and consumption has moved by at most $M_k t \leq M_k h/2$), so by soundness $|\partial_r c_k| \leq \max(|U_k|, |L_k|) < M_k$ on the zone at r, one-sidedly at kinks. The one-sided derivatives at $r_0 \pm t$ are strictly below M_k, and the derivative extends continuously to the closed pieces, so there is $\varepsilon > 0$ with $|\partial_r c_k| < M_k$ on $|r - r_0| < t + \varepsilon$ off the kinks; by absolute continuity $|c_k(r) - c_k(r_0)| \leq \int |\partial_r c_k| \leq M_k|r - r_0|$ there, so $t + \varepsilon/2 \in T$. A closed subset of $[0, h/2]$ that contains 0 and has no largest element below $h/2$ is all of $[0, h/2]$. □

The hypotheses of the lemma are the regularity of Section 4 and the soundness of the recursion, which is Section 7 read with the cell's rate range in place of a rate. The lemma is the one piece of the construction that is neither a formalised theorem nor arithmetic: it is a paper argument, and a short one.

8.2 What makes it run

Three refinements were needed, and each failed version is recorded in the companion.

The zone. The assumption (7) cannot be made on the whole grid: near the grid's top the midpoint bounds U^0 are enormous and the transported widths with them, and a few grid points just below the top then fail (8). The assumption is confined to the zone the recursion actually reads from the cohort's region, $Z_{J-1} = [0, 1.2\rho + 2]$ at the newborn stage and $Z_k = [0, \max(1.2\rho_{\text{age}} + 2,\ (1 - \kappa_{k+1})(y_{\max} + R \sup Z_{k+1}) + 2)]$ working backward, which is closed under the saving map by the propensity floor; its top runs from about 32 at the newborn stage through 137 at stage thirty to 2 at the last stage. The recursion skips intervals above the zone, where the fallback bounds would make the rationals enormous without being read.

The floor μ. Near the corner the midpoint bound U^0 is of order a, tiny, and the cell pass — which cannot see that the household is constrained at every rate in the cell — returns a bound above λU^0. The additive floor $\mu = 1$ covers those states.

The multiplier λ does not help. Cells of 0.02 percentage points fail (8) by nine percent at every λ tried, because the transported width scales with λ and so does the bound it produces; cells of 0.1 points fail it forty times over. Cells of 0.01 points verify at a ratio of 0.75, and cells of 0.005 at 0.63. The width is what it is: 0.01 percentage points at the present enclosure quality.

The cost of the cell is the *inflation* of B_d over the midpoint's: at 5% the cell bound is 6.9 against a midpoint bound of 6.3, nine percent; at 6%, 8.1 against 7.4. The transport does not spoil the margin.

9 Covering the bracket

The bracket $[2.3\%, 6\%]$ is partitioned into 370 cells of 0.01 percentage points, run in six batches: $[4\%, 6\%]$ first, then five extensions downward to 2.3%, where the Lean theorem of Section 12 takes over. For each cell the certificate runs the point enclosure at the midpoint (about twenty-four minutes on the fine grid), the derivative recursion at the midpoint, the transported enclosure and the cell recursion, and the verification (8); a cell that failed would be split in two and both halves rerun, down to a minimum width of 0.001 points. Each batch is resumable, with one line per finished cell in `numerics/rational/runs/bracket_g3_*.txt`; the whole took about 470 processor-hours on seven cores, at 76 minutes per cell.

No cell needed splitting. Every cell certified at its first width, with the verification ratio $\max |U|/M$ about three quarters on every one of them, and the certified bounds are:

sub-range	$\max B_d$	$\min \lvert D' \rvert$	worst $B_d/\lvert D' \rvert$	cells
$[2.3\%, 3.0\%]$	5.10	46.49	0.110	70
$[3.0\%, 4.0\%]$	6.45	39.06	0.165	100
$[4.0\%, 5.0\%]$	7.82	33.28	0.235	100
$[5.0\%, 6.0\%]$	9.15	28.70	0.319	100

B_d rises smoothly from 4.28 at the bottom of the bracket to 9.15 at the top, while $|D'|$ falls, so the worst ratio, 0.319, sits at the top cell $[5.99\%, 6.00\%]$. On every cell $B_d < \min_{\text{cell}} |D'|$, with a margin of at least three. By Lemma 4.1 applied cell by cell, for any two rates $r_1 < r_2$ in the same cell,

$$E(r_2) - E(r_1) \;\geq\; (r_2 - r_1)\big(\min_{\text{cell}} |D'| - B_d\big) \;>\; 0,$$

and since adjacent cells share an endpoint, E is strictly increasing on the whole bracket.

10 The bracket contains an equilibrium

Strict monotonicity of E gives at most one zero. For exactly one, E must change sign on the bracket, and the same enclosures certify that it does (`numerics/rational/levelcert.py`). Run the cohort-assets recursion of Section 2 with the *interval* policy. At a grid point a_i and state z the saving $g_{k+1}(a_i, z)$ lies in $[m(a_i) - \overline{c}_{k+1}(a_i, z),\ m(a_i) - \underline{c}_{k+1}(a_i, z)]^+$ by the enclosure, with $m(a_i)$ rounded down in the first and up in the second; A_k is nondecreasing in assets, because saving is and the recursion preserves monotone functions; so a lower bound on $A_{k+1}(a_i, z)$ is a_i plus the π-expectation of the lower bound on A_k read at the grid point *below* the lower bound on saving, and an upper bound is the same with upper bounds and the grid point *above*, each rounded outward. Sixty steps from $A_0 = a$ give $S(r) \in [\underline{S}, \overline{S}]$:

r	$S(r)$ certified	$D(r)$	verdict
4%	[3.821, 3.966]	4.6875	$S < D$
6%	[4.950, 5.159]	4.0179	$S > D$

The widths are the grid rounding of sixty compositions and are of no consequence against gaps of 0.72 and 0.93. So $E(4\%) < 0 < E(6\%)$.

11 Proof of Theorem 2.1

S is continuous on $[2.3\%, 6\%]$ (`continuousOn_olgSupply`); the hypotheses of that theorem are the stage properties on a reachable family, and the enclosure's reach, $\rho_j \leq 44$, is such a family for any cap above it, with cap slack certified by the enclosure itself (the upper bound on saving never approaches the grid's top). D is continuous. By Section 10, $S(4\%) < D(4\%)$ and $S(6\%) > D(6\%)$, so by the intermediate value theorem (`exists_olgEquilibrium`) there is a rate in $(4\%, 6\%)$ with $S = D$. By Section 9, $E = S - D$ is strictly increasing on $[2.3\%, 6\%]$, so that rate is the only one in $[2.3\%, 6\%]$. By Theorem 12.1 of Section 12, $S < D$ at every rate in $(-8\%, 2.3\%]$, so there is none there. □

Read backwards, the proof uses: the certified endpoint levels (Section 10, arithmetic on the enclosures); the certified slope bound on every cell (Sections 7 and 8, arithmetic on the enclosures plus Lemma 8.1); the enclosures themselves (Section 6, instances of the interval Euler step); Lemma 4.1 and the derivative formula (6) (paper); and the companion's theorems of Section 3 (kernel-checked).

12 Outside the bracket, by proof term

Below and above the bracket the question changes character. Near the equilibrium only the slope certificate can separate supply from demand, because the proved level bounds are too loose there; away from it the level bounds of the companion suffice, and they are cheap enough to evaluate inside Lean on whole ranges of rates.

The two bounds. The companion proves an affine ceiling on accumulated assets, $A_k(a, z) \leq \bar{F}_k(z) + G_k a$ with $\bar{F}$ an explicit recursion in κ_k, the cohort slope G_k and a risk-adjusted human wealth built from power means of exponent $-\gamma$, valid whenever the cap is slack everywhere (`olgCapital_le_cohortCeil`); and a clamped floor, the recursion $\Lambda_0(z, a) = a$, $\Lambda_{k+1}(z, a) = a + \sum_{z'} \pi(z, z')\Lambda_k(z', \Phi_{k+1}(z, a))$ with Φ the saving floor $\max\{0, (1-\kappa_k)(y_z + Ra) - \kappa_k H_k(z)\}$, valid whenever a lifetime of feasible saving stays under the cap (`le_olgCapital_at`). Supply is at most the ν-average of $\bar{F}_{59}$ and at least that of $\Lambda_{59}(\cdot, 0)$, each over sixty.

From a rate to a cell. Every ingredient is monotone in the rate R or in the patience factor $\text{þ} = (\beta R)^{1/3}$: κ_k rises with R and falls with þ, human wealth falls with R, the cohort slope rises with R and falls with κ, and the power mean is monotone in its arguments. On a cell $[r_1, r_2]$ a checker (`EnclCert/Outer.lean`) brackets þ by rational cube-root bounds, runs each recursion with every ingredient taken at the end of the cell that makes the bound larger, rounding outward over $D = 10^{12}$, and compares the ceiling with demand at r_2 or the floor with demand at r_1. The floor is carried on a grid of 650 asset levels and read at the grid point below the certified saving floor, which is sound because Λ_k is monotone in assets.

Every stored value is re-verified by an exact integer inequality with denominators cleared, and only those inequalities are used by the bridge (`OLG/OuterBridge.lean`), which proves by induction on the stage that each stored number bounds the real quantity it stands for, discharges the slackness hypothesis of the ceiling from impatience at every stage (the limiting propensity $1 - \text{þ}/R$ where $\text{þ} < R$, and $y_{\max} + r\bar{a} \le 0$ at rates negative enough), and checks the reach hypothesis of the floor at the top of the cell. The cap is 10^6, which the reach at 20% needs; the ceiling side is indifferent to it.

The meshes. The ceiling crosses demand near 2.35% and the floor near 8.4% at this calibration. The low mesh is 165 cells from −8% to 2.3%, of 0.1 points to 1.5%, 0.02 to 2.1% and 0.005 to 2.3%; the high mesh is 121 cells from 8.7% to 20%, of 0.01 points to 9.5%, 0.1 to 12% and 0.5 to 20%. Both verdicts are established by `native_decide` (`lowCheck_eq_true`, `highCheck_eq_true`), about four minutes of compiled arithmetic each; a covering lemma puts every rate of a range in some cell of an increasing mesh.

Theorem 12.1 (No equilibrium outside $[2.3\%, 8.7\%]$)**.** *At $\gamma = 3$ and the calibration of Section 2 with cap 10^6, capital supply is strictly below demand at every rate in $(-8\%, 2.3\%]$ and strictly above it at every rate in $[8.7\%, 20\%]$.*
Formal statement: *`EnclCert.olg7_supply_lt_demand_low, EnclCert.olg7_demand_lt_supply_high`.*

Unlike Theorem 2.1, this one is a Lean theorem: its only axiom beyond classical Mathlib is the one `native_decide` introduces for the two compiled verdicts. What it leaves uncovered below 20% is $(6\%, 8.7\%]$, where neither bound separates supply from demand and only the slope certificate of Sections 7–9 can; extending that certificate to it is about 270 further cells of the same run. Above 20% the formal model's cap starts to bind for a household that saves everything feasible, and the capped economy stops being the one of interest.

13 Standing, and what remains

13.1 What kind of result this is

step	standing
stage properties, propensity floor, cohort slope, earnings marginal, cohort envelope, continuity, existence, interval Euler step, chord and secants	theorems, checked by the Lean kernel
the derivative formula (6); Lemma 4.1; Lemma 8.1; piecewise-C^1 regularity of the policies in the rate	paper arguments, given above
the enclosures at two hundred midpoints and two endpoints; the derivative recursion on each cell; the verification (8); the endpoint levels	exact rational arithmetic, rounded outward, evaluated by `derivcert.py` and `levelcert.py`

The arithmetic row is where the standing is weakest, and it is weak in a specific way: each inequality the programs check is an instance of a formalised theorem (the enclosure) or of a recursion written out in Sections 7–10 (the derivative bound, the level bound), but the programs that evaluate them were not proved to evaluate what this note says they evaluate. The companion has already removed that gap once, for the pairwise enclosure certificate at $(4\%, 4.6\%)$: a Lean checker over naturals, its verdict established by `native_decide` in a quarter of an hour, and a bridge from the checker's inequalities to the theorems behind them (`enclCheck_eq_true`, `olg7_increment`). The derivative certificate has the same shape, with three additions: the interval recursion of Section 7 as a checker, the level recursion of Section 10, and Lemma 8.1 as a theorem, the one genuinely new piece of mathematics. At seventy-six minutes per cell in Python, three hundred and seventy cells in compiled Lean is a weekend of machine time, not a new argument.

13.2 What the theorem does not say

It is one calibration: one chain, one β, one γ, one technology. Nothing transfers to a neighbouring calibration except by rerunning the certificate, which is the price of a certificate over a proof. It is one bracket, nearly: Theorem 12.1 excludes every rate in $(-8\%, 2.3\%]$ and in $[8.7\%, 20\%]$ by proof term, so what remains open below 20% is $[2.3\%, 4\%)$ and $(6\%, 8.7\%]$, where the slope certificate has not yet been run; until it is, the honest statement is exactly one equilibrium on $(-8\%, 6\%]$ and none in $[8.7\%, 20\%]$. And it is $\gamma = 3$: the same run at $\gamma = 5$, where the single-rate bound is 12.8 against 39 at 4%, is the same command with one argument changed and about the same machine time, but it has not been run.

13.3 What would raise it

In order of return on effort. A *transported* derivative bound in place of the constant multiple λU^0 in (7) — carrying the assumption through the recursion rather than freezing it — should admit cells three to five times wider, since the verification ratio is pinned at 0.75 by the constant multiple and not by the margin. The $\gamma = 5$ bracket, which turns the theorem into a statement about Aiyagari's second calibration. The Lean replay of the derivative checker with Lemma 8.1, which turns the certificate into a theorem with one `native_decide` axiom. And the formalisation of piecewise analyticity of the stage policies in the rate, which the regularity hypotheses of Lemmas 4.1 and 8.1 rest on and which is at present a paper argument from the analytic implicit function theorem.

A Reproduction

All files are under `WriteUps/WriteUpOLG/numerics/rational/` in the repository.

- The bracket runs (resumable, results appended to the file named), with the environment `ENCL_FINE=fine ENCL_NOCACHE=1 ENCL_MU=1 ENCL_LAMS=2 ENCL_PROCS=7 ENCL_RESULTS=runs/bracket_g3_<name>.txt` and the arguments `derivcert.py 3`

`r_lo r_hi n 1 fixed`: `0.04 0.06 200` (4to6), `0.037 0.04 30` (37to40), `0.031 0.037 60` (31to37), `0.0275 0.031 35` (275to31), `0.0233 0.0275 42` (233to275), `0.023 0.0233 3` (230to233).

- A batch's summary (leaves, splits, worst margin, coverage): `python3 runs/bracket_summary.py runs/bracket_g3_4to6.txt 0.04 0.06`.
- A single rate, for the first table of Section 7: `ENCL_FINE=fine python3 derivcert.py 3 0.04 0.04 0`.
- The endpoint levels: `ENCL_FINE=fine python3 levelcert.py 3 1/25` and `... 3 3/50`.
- The derivative-form measurements on the solved model, which motivated the weighted recursion: `numerics/slopebnd.m`, `slopebnd2.m`, `dcdr.m`.

The calibration is in the header of `derivcert.py`: $J = 60$, $n_z = 7$, $\beta = 24/25$, $\rho = 9/10$, $\alpha = 9/25$, $\delta = 2/25$, the Rouwenhorst chain, the seven earnings levels and their binomial invariant law, the grid, and $D = 10^{15}$.

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